

EDUCATION

First Class BSc (Hons) Artificial Intelligence and Data Science | Information Institute of Technology (IIT), Sri Lanka
affiliated with Robert Gordon University (RGU), Scotland

June 2024

PROGRAMMING SKILLS

- Languages: Python, R, Java, JavaScript, HTML
- Database/Server: My SQL, MongoDB
- Frameworks: Hadoop, Spark, TensorFlow, Pytorch, Scikit-learn, Pandas, Seaborn, NumPy, NLTK, SpaCy, Node.js, Lime, Hugging Face, Lang Chain, LlamaIndex ,Guidance AI, Deap library for optimization
- Cloud Platforms and services: AWS: Amazon Sage Maker, AWS Lambda, Amazon Event Bridge, and Amazon ground truth labeling, Amazon bedrock

INDUSTRY EXPERIENCE

Machine Learning Engineer, Ascentic Pvt Ltd, Sri Lanka

May 2024-Present

- Implemented **Retrieval-Augmented Generation (RAG) pipelines** using GPT-3.5 for multilingual product categorization; optimized prompt engineering strategies to enhance classification accuracy.
- Engineered preprocessing pipelines for 70M+ **PostgreSQL records** using Python (Pandas, JSON, Regex) and DBEaver; and delivered **structured datasets from complex nested JSONs across 3+ client projects** for enabling ML model training
- Performed batch inference with Mistral, Claude 3.7 for PDF extraction and categorization using **Amazon bedrock**.
- Contributed to financial dimension prediction by collaborating with finance teams for rule mining and feature extraction.
- Developed **cross-functional annotation workflows** for invoice PDF extraction and categorization; coordinated between machine learning and annotation teams to generate stratified test sets, verify label quality, and optimize dataset preparation pipelines.

Data Science Intern, Ascentic Pvt Ltd, Sri Lanka

May 2022 – August 2023

- **Collaborated on the Qvalia product categorization project**, developing deep learning methodologies for hierarchical retail taxonomy classification and improving language model performance.
- Experimented with various machine learning models, sequential models, and **fine-tuned pre-trained models** such as BERT for language classification tasks.
- Conducted a comprehensive literature review on retail domain research projects and **successfully presented machine learning proof-of-concept (POC)** work to internal teams and external clients.
- Performed data retraining for multi-class product categorization using **AWS SageMaker** and explored AWS services such as S3, Textract, and Amazon Translate to support ML pipeline automation.

ACADEMIC RESEARCH EXPERIENCE

Prompt-based approach for improving sustainability based on food, location, and customer service in hotels with

Senior Lecturer Prasan | Grade A

July 2023-May 2024

- Developed a **novel ensemble prompting technique** for improving aspect-level sentiment summarization, including divergent thinking and "what if" strategy prompts for sustainable strategy formulation using Hugging Face, Guidance AI, and LangChain
- Selected and evaluated open-source 7B models (**LLaMA, Zephyr, Mistral**) and closed-source models (**GPT-3.5**) as baselines.
- Built a prototype website using React.js, Node.js, and Flask for demonstration.

Data Science Group Project with Senior Lecturer, Miss Fathima Farath

September 2021- April 2022

- Led a team of 4 to develop an innovative web and mobile application for diagnosing COVID-19 using respiratory sounds and voice to self-monitor patient progress.

- Designed and pre-processed audio datasets, created, tested, and **ensembled 6 CNN** models for positive case detection using Python, Librosa, and TensorFlow.

HIGHLIGHTED PROJECTS

Profit, Lead time, and Route planning optimization in E-commerce

Nov 2023 -December 2023

- Led a group 10 in managing and designing the project timeline; worked on route optimization and website design with React and Node.js.
- Utilized Non-Dominated Sorting Genetic Algorithm II for **multi-objective optimization** (distance, lead time, and profit) using an e-commerce company's supply chain data with Deap library

Brain Tumor detection with explainable AI and a chatbot for assisting medical students

July 2023 -September 2023

- Collaborated with a team of 3 on detecting brain tumor types and segmenting the brain with pre-trained models such as Efficient Net and Mobile Net with **explainability with LIME**.
- Created a vector store of relevant research case reports on brain tumors (Retrieval-Augmented generation) and prompt tuned GPT 3.5 turbo for assisting medical students and doctors on new treatments for brain tumors

Accident and Violence Detection at the Edge

March 2023-April 2023

- Led a group of 4 members to develop a mobile application to run a CNN model on the edge. The application is capable of inferencing at the end device and retraining the model in the cloud using **AWS Lambda, Amazon Event Bridge, Amazon Sage Maker, and Amazon ground truth labeling**.
- Utilized TensorFlow and TensorFlow Lite to implement CNN architecture and make it scalable.

Identifying path using search techniques in 8X8 matrix

Dec 2021

- Implemented Search technique (Informed and Uninformed) solutions for path in 8X8 matrix in Python and Java.

KEY ACHIEVEMENTS

Poster presentation in Cutting Edge (Project Showcasing Event) at IIT

September 2023

- Led a team of 3 and presented a project on Brain Tumor detection with explainable AI and a fine tuned chatbot for assisting medical students

Participant in a team in Datathon

August 2023

- Site selection for renewable energy generation in Sri Lanka using machine learning algorithms. Utilized k-means clustering techniques, SVM, decision trees, and Artificial Neural networks.

2nd Runner Up as a Team in TADHack 2021

September 2021

- Mobile and websites were implemented to diagnose heart attacks and blood pressure with MERN stack and tensorflow

The annual prize-giving recipient, Belvoir College International

- Developed an Award for **Excellence in IAL Mathematics**
- Awarded **Excellence for 3A*s AND A (IGCSE)** in the annual prize-giving

COMMUNITY INVOLVEMENT

Knowledge sharing session, Ascentic, volunteer

September 2022 -December 2024

- Conducted internal company sessions on Python (exception handling), prompt engineering, and Generative AI.
- Delivered a presentation on internship experiences to ~50 university students at General Sir John Kotelawala Defence University (KDU).

Reviewer – IEEE conference ASONAM 2024

May 2024

- Reviewed a conference submission and provided detailed evaluation and feedback as part of the peer review process for IEEE ASONAM (**16th International Conference on Advances in Social Networks Analysis and Mining**) 2024.

We can project, Rotaract club, chairperson

Jan 2021-April 2021

- Led a community service project focusing on young women's sanitation, recognized as the **Most Outstanding Project**.